QUANTITATIVE LOCAL MODELS FOR HERMITIAN–EINSTEIN METRICS

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Abstract. We give a quantitative local model for Hermitian–Einstein metrics near a fixed locally free polystable bundle on a compact Kähler manifold. Retaining the full convergent Kuranishi obstruction, we compare every small endomorphism Dolbeault eigenvalue with the commutator form of the automorphism action on harmonic deformations, and the harmonic metric on smooth pieces of the reduced moduli space with the flat quotient metric. The model reconstructs every sufficiently nearby Hermitian–Einstein solution. The comparisons are relative and uniform through stabilizer changes and arbitrary separation of positive eigenvalues of the finite form. Canonical harmonic coordinates give quadratic error at fixed polarization, with the same order for varying polarizations in moving harmonic backgrounds. The proof derives the geometric estimates for finite balancing and complementary elliptic reduction from orthogonality of the quadratic reconstruction and gauge terms. We construct a stable rank-two bundle on a smooth bicubic Calabi–Yau threefold with essential cubic deformation equations. The theorem controls all nearby polystable deformations of the direct sum of three copies, including families with arbitrarily separated small positive eigenvalues.

1. Introduction

The local deformation theory of a polystable holomorphic bundle is finite dimensional, whereas its Hermitian–Einstein metric is obtained from a nonlinear elliptic equation. Near a bundle with additional automorphisms, the linearized equation develops small eigenvalues. Several of these eigenvalues can tend to zero at unrelated rates. An absolute error estimate for the curvature or the operator then gives no relative information about the weakest directions. The same issue arises when comparing the harmonic metric on bundle moduli with a finite-dimensional quotient metric.

We prove that canonical harmonic Kuranishi coordinates provide one finite-dimensional model for all three comparisons near any fixed locally free polystable compact-Kähler background. No formality hypothesis is imposed. The complete convergent obstruction is retained, and the estimates remain uniform when the stabilizer type changes. The essential geometric observation is that the quadratic terms of both the Kuranishi reconstruction and the complementary Hermitian gauge are orthogonal to harmonic one-forms. This gives a quadratic estimate for the ambient symplectic form and for the complementary quotient form. The finite balancing theorem of [Wue26h, Theorem 2.1], the relative spectral reduction, and minimization over gauge directions then apply on the same entire local family.

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1.1. **The finite data.** Let (X, ω_0) be a compact connected Kähler manifold of complex dimension n, and fix a locally free polystable bundle

$$E_0 = \bigoplus_{i=1}^{\ell} F_i \otimes W_i \tag{1}$$

with pairwise nonisomorphic stable factors of equal slope. Put $r_i = \operatorname{rank} F_i$, $m_i = \dim W_i$, and $R = \sum_i r_i m_i$. Fix a direct-sum Hermitian–Einstein metric h_0. Let ω_t be a smooth finite-dimensional family of Kähler forms, with ω_0 its central value. We fix the determinant holomorphic structure. Degrees and slopes are normalized by

$$\deg_\omega F = \int_X c_1(F) \wedge \frac{\omega^{n-1}}{(n-1)!}, \qquad \mu_\omega(F) = \frac{\deg_\omega F}{\operatorname{rank} F}.$$

Write $d = \bar\partial_0$ on $\operatorname{End} E_0$, and use the unnormalized (h_0, ω_0) inner products. Denote the harmonic projections and Green operators by H_q and G_q.

The harmonic space $V = H^1(X, \operatorname{End} E_0)$ is the full Ext-quiver representation of $G = \prod_i \mathrm{GL}(W_i)$; its compact form is $K = \prod_i \mathrm{U}(W_i)$. Orthonormal harmonic bases in each $\mathrm{Ext}^1(F_i, F_j)$ give the arrow coordinates x_a. Loops and all multiplicities are included. Set

$$V_{\det} = \ker\bigl(\operatorname{tr} : V \longrightarrow H^1(X, \mathcal{O}_X)\bigr).$$

For small $x \in V$ the canonical harmonic recursion defines

$$a(x) = x - d^* G_2(a(x) \wedge a(x)), \qquad \kappa(x) = H_2(a(x) \wedge a(x)). \tag{2}$$

Put $Z = \kappa^{-1}(0) \cap V_{\det}$ and $E_x = (E_0, \bar\partial_0 + a(x))$ for $x \in Z$. The full analytic structure of Z is relevant to deformation theory. For quotient metrics we use its underlying reduced analytic space.

The flat moment map and central target are

$$\begin{aligned} \mu_i(x) &= \sum_{h(a)=i} x_a x_a^* - \sum_{t(a)=i} x_a^* x_a, \\ \zeta_i(t) &= 2\pi\bigl(r_i \mu_{\omega_t}(E_0) - \deg_{\omega_t} F_i\bigr), \qquad \sum_i m_i \zeta_i(t) = 0. \end{aligned} \tag{3}$$

Here $h(a)$ and $t(a)$ denote the head and tail of an arrow. The ordinary multiplicity trace pairs the moment covector with Hermitian matrices. The spectral mass has different rank weights:

$$\begin{aligned} \mathfrak{g}_0 &= \left\{ (\xi_i) \in \bigoplus_i \operatorname{End} W_i : \sum_i r_i \operatorname{tr} \xi_i = 0 \right\}, \qquad C_x \xi = [\xi, x], \\ q_x(\xi) &= \|C_x \xi\|^2, \qquad \|\xi\|_0^2 = \operatorname{Vol}(\omega_0) \sum_i r_i \operatorname{tr}(\xi_i^* \xi_i). \end{aligned} \tag{4}$$

Let $\nu_1(x) \le \cdots \le \nu_N(x)$ be its eigenvalues, with complex multiplicity and zeros included, where $N = \sum_i m_i^2 - 1$.

Locally near the origin consider

$$\mathcal{Q}_t = (Z \cap \mu^{-1}(\zeta(t)))/K.$$

Its neighborhoods are images of small balanced neighborhoods; a bounded ball is not declared invariant under G. Refine the reduced complex quotient into smooth local pieces with

locally constant stabilizer type and smooth lifts, refining also at analytic singularities. Their existence is explained in Section 3.4. On a tangent represented by a balanced lift v define

$$g_{\mathrm{fin},x}([v],[v]) = \inf_{\xi\in\mathfrak{g}_0} \|v - C_x\xi\|^2. \tag{5}$$

This is the flat Kähler quotient metric restricted to the chosen piece. The tangents in this formula are genuine tangents to those pieces, not arbitrary Zariski tangents of a singular coarse space.

1.2. **The local comparison theorem.** Let b_t be the Rth root of the ratio of the normalized ω_t-Hermitian–Einstein determinant metric to $\det h_0$. We normalize it by $\int_X \log b_t\, dV_0 = 0$, where $dV_0 = \omega_0^n/n!$. Then $b_0 = 1$ and $\|\log b_t\|_{C^j} \le C_j|t|$.

Theorem 1.1. *For the fixed data above, there are a parameter neighborhood, nested coefficient neighborhoods, and an open neighborhood of the central actual Hermitian–Yang–Mills class with the following properties. Write $\delta = |t| + \|x\|^2$ and shrink the neighborhoods so that $C\delta < 1/2$.*

(1) *Every $x \in Z \cap \mu^{-1}(\zeta(t))$ in the inner neighborhood has a Hermitian–Einstein metric with the prescribed normalized determinant and*

$$e^{-C\delta} b_t h_0 \le h^{\mathrm{HE}}_{t,x} \le e^{C\delta} b_t h_0. \tag{6}$$

The construction identifies neighborhoods in $\mathcal{Q}_t$ and the actual fixed-determinant HYM quotient, uniformly for small t, and covers every sufficiently nearby actual solution. Metrics are unique modulo determinant-preserving holomorphic automorphisms; the unitary HE class is unique.

(2) *The Dolbeault Laplacian on $\operatorname{End}_0 E_x$, for $h^{\mathrm{HE}}_{t,x}$ and ω_t, has exactly N eigenvalues below a fixed $\gamma > 0$, and*

$$(1 - C\delta)\nu_j(x) \le \lambda_j(t,x) \le (1 + C\delta)\nu_j(x) \quad (1 \le j \le N). \tag{7}$$

Its kernel is naturally identified with $\ker C_x$, and its next eigenvalue is at least γ. For full $\operatorname{End} E_x$, add the one exact scalar zero; all other scalar modes have a uniform positive gap.

(3) *The local identification f_t is a homeomorphism, preserves the local complex-orbit description, and is smooth on the smooth pieces specified above. Their harmonic Kodaira–Spencer metric satisfies*

$$(1 - C\delta) g_{\mathrm{fin}} \le f_t^* g_t^{L^2} \le (1 + C\delta) g_{\mathrm{fin}}. \tag{8}$$

The same comparison constant works through changes of stabilizer.

Constants and radii may depend on the fixed background, ranks, multiplicities, harmonic dimensions, positive background elliptic gaps, Sobolev and elliptic constants, the selected regularity order, and bounds for the given family ω_t. They are independent of every positive $\nu_j(x)$, ratios of positive finite eigenvalues, positive singular values of C_x or $D\kappa_x$, and stabilizer type.

For $N = 0$ and $R > 1$, the trace-free assertion is a uniform gap on the entire zero-form spectrum. For $R = 1$, $\operatorname{End}_0 E_0 = 0$ and that assertion is vacuous; the full endomorphism statement retains its scalar zero. The theorem asserts no uniform radius for separate charts on smooth pieces, nor a continuous tensor on untransported tangent vectors when their dimension changes.

The error can be sharpened when the harmonic coordinates move with the polarization. Continue the HE metric of each fixed stable factor smoothly to $h_i(t)$ and use the harmonic

construction for $h_{\mathrm{gr},t} = \bigoplus_i h_i(t) \otimes I_{W_i}$ and ω_t. Proposition 5.1 proves all three comparisons with error $C\|x\|_t^2$, relative to that background and its spectral mass. The split bundle need not remain polystable for $t \neq 0$. This distinction preserves the stronger rates of the earlier extension theorems when a polarization moves predominantly along a stability wall.

1.3. **An application with cubic obstruction.** We construct a stable rank-two bundle A, with trivial determinant, on a smooth bicubic threefold in $\mathbb{P}^2 \times \mathbb{P}^2$, whose analytic deformation germ is

$$\operatorname{Specan} \mathbb{C}\{a, b\}/(a^2 b, ab^2). \tag{9}$$

Its quadratic obstruction vanishes, but mixed directions are obstructed at order three. Section 6 proves smoothness of the compact geometry, stability, and equality of the complete curve and bundle deformation germs. The construction modifies the bicubic Serre method of Thomas [Tho99]; its deformation equations differ from his corrected quadratic example.

Theorem 1.1 applies to the entire 18-dimensional harmonic deformation problem at $A \otimes \mathbb{C}^3$, retaining its full matrix-valued obstruction. The trace-free finite algebra has dimension eight. An explicit diagonal subfamily has two eigenvalues on an arbitrarily weak positive scale and four on a larger scale, with changing exact kernel dimensions. This subfamily illustrates spectral uniformity; the application concerns the whole local moduli problem. The reduced space of (9) is the node $ab = 0$. The harmonic metric is defined on reduced smooth pieces and does not measure its embedded nilpotent structure.

1.4. **Relation to earlier work and proof strategy.** The Donaldson–Uhlenbeck–Yau correspondence gives the qualitative existence of HE metrics [Don85, UY86]; see also [Kob87]. Buchdahl–Schumacher develop the equivariant Kuranishi and automorphism description [BS22]. Their Proposition 6.2 and Lemma 6.3 already give a quadratic complementary correction and a quartic finite moment remainder. Their Theorems 7.2 and 8.1 provide quantitative raywise correction and common existence neighborhoods. The present comparison requires relative control in each weak orbit direction, which does not follow from an absolute quartic remainder. Their treatment of harmonic moduli metrics also includes obstructed spaces under the hypotheses of [BS21, Theorem 11]. Delloque's local correspondence supplies the common-parameter slices, morphism description and quotient topology used here [Del25, Section 3.2, Propositions 4.3, 4.5, 5.6, Corollary 5.2 and Theorem 7.4]. Convergent Ext-quiver relations are also part of Toda's local description of sheaf moduli [Tod18, Theorem 1.1].

Several quantitative mechanisms are available in earlier special geometries. The complete multiplicity algebra and relative spectral reduction occur in [Wue26b]; complementary minimization and nonabelian quotient estimates occur in [Wue26d, Wue26e]. The finite balancing theorem of [Wue26h] already applies to arbitrary quivers and locally invariant analytic subsets. The additional result here is the general geometric verification that permits these mechanisms to act simultaneously on the entire convergent Kuranishi family, without formality or a positive scale ratio bound. The cubic compact example provides a locally free application beyond the earlier extension and K3 settings.

The proof begins with the two orthogonal quadratic jets

$$-d^* G_2(x \wedge x), \qquad -d\sigma_2(x).$$

Their pairing with harmonic forms vanishes, giving the quadratic ambient estimate. Finite balancing then solves the remaining moment equation with a small parallel gauge. For spectra, the distinct identity $D_a \xi = -D_x a(C_x \xi)$ controls the off-diagonal Schur term by the finite

energy of the same vector. For quotient metrics, minimizing first over the uniformly coercive infinite-dimensional complement gives a positive form close to the harmonic form; only then is the finite orbit minimum taken. Derivatives of the balancing gauge are vertical and disappear. These separate arguments explain why one relative constant survives arbitrarily weak finite directions.

2. Canonical harmonic coordinates

We use the full endomorphism complex until imposing the determinant equation. At the HE background, the holomorphic zero-forms are parallel, so

$$P = H^0(X, \operatorname{End} E_0) = \bigoplus_i I_{F_i} \otimes \operatorname{End} W_i.$$

Its trace-free part is identified with $\mathfrak{g}_0$ in (4). Write Q_0 for the orthogonal complement of $P_0 = P \cap \operatorname{End}_0 E_0$ in trace-free zero-forms, and Q_H for the corresponding projection on Hermitian trace-free zero-forms. All complements in this section are fixed at (h_0, ω_0). Choose an integer $k > n + 2$ and use $W^{k,2}$ for one-forms and $W^{k+1,2}$ for complex gauges.

2.1. **Convergent reconstruction and exact equivariance.**

Lemma 2.1. *The recursion (2) has a unique analytic solution on a small ball in V, smooth in the spatial variables. Writing $r = \|x\|$ and $J_x = D_x a$, one has, in every fixed C^j norm,*

$$\begin{gathered} a = x + O(r^2), \qquad J_x = \iota + O(r), \\ H_1 a = x, \qquad d^* a = 0, \qquad H_1 J_x = I, \qquad d^* J_x = 0. \end{gathered} \tag{10}$$

Here ι is harmonic inclusion. The reconstructed operator is integrable exactly when $\kappa(x) = 0$, and $\operatorname{tr} a = \operatorname{tr} x$. The maps are locally G-equivariant. With $D_a = d + [a, \cdot]$ on endomorphism forms,

$$D_a \xi = -J_x C_x \xi \qquad (\xi \in P). \tag{11}$$

The identity holds on the whole ambient ball.

Proof. Sobolev multiplication and the bounded map $d^* G_2 : W^{k,2} \to W^{k+1,2}$ make the right side of (2) a contraction on a small ball. Analytic parameter differentiation gives the first two estimates. The nonlinear term is coexact, proving the harmonic and co-closed identities. Bootstrapping the equation gives spatial smoothness and the estimates in any specified finite regularity order.

Put $\mathcal{R} = da + a \wedge a$. Hodge decomposition gives

$$\mathcal{R} = H_2(a \wedge a) + d^* d G_2(a \wedge a), \qquad d^* \mathcal{R} = 0, \qquad H_2 \mathcal{R} = \kappa.$$

The Bianchi identity is $d\mathcal{R} + [a, \mathcal{R}] = 0$. When $\kappa = 0$, Hodge decomposition of $\mathcal{R}$ therefore implies

$$\mathcal{R} = d^* G_3 d\mathcal{R}, \qquad \|\mathcal{R}\|_{W^{k-1,2}} \le C r \|\mathcal{R}\|_{W^{k-1,2}}.$$

For sufficiently small r this forces $\mathcal{R} = 0$. In complex dimension less than three the degree-three term is absent, with the same conclusion. Conversely, integrability implies $\kappa = 0$ by harmonic projection. Since $\operatorname{tr}(a \wedge a) = 0$ and trace commutes with the scalar Hodge operators, $\operatorname{tr} a = \operatorname{tr} x$.

Every element of G is parallel. Its action commutes with the Hodge operators, even though the action need not be unitary. Uniqueness of the contraction gives $a(gx) = g a(x) g^{-1}$ whenever the local expressions are defined. Differentiation gives $J_x[\xi, x] = [\xi, a] = -D_a \xi$, proving (11). □

In particular the entire scalar harmonic deformation is removed by $V_{\det}$, also when $H^1(X,\mathcal{O}_X)\neq 0$. This subspace need not itself be a full quiver representation. We apply finite balancing on the full representation V and impose $V_{\det}$ as an additional invariant analytic equation. Neither the construction nor any later estimate replaces κ by its quadratic term.

2.2. **The complementary Hermitian–Einstein equation.** On the fixed Hermitian bundle define the curvature moment with fixed density $dV_0=\omega_0^n/n!$ by

$$\mathcal{M}_t(A)=\frac{iF_A^0\wedge\omega_t^{n-1}}{(n-1)!\,dV_0}. \tag{12}$$

The superscript denotes trace-free curvature. A possibly nonintegrable $(0,1)$-operator determines a unitary connection by its adjoint $(1,0)$-part, so the following equation is defined on the ambient ball.

Lemma 2.2. *There is a unique small $\sigma(t,x)\in Q_H W^{k+1,2}(\operatorname{Herm}_0 E_0)$ satisfying*

$$Q_H\mathcal{M}_t\big(e^{\sigma(t,x)}\cdot(\bar\partial_0+a(x))\big)=0. \tag{13}$$

It is smooth and K-equivariant, and

$$\|\sigma(t,x)\|_{C^j}\le C_j(|t|+r^2),\qquad \|D_x\sigma(0,x)\|_{V\to C^j}\le C_j r. \tag{14}$$

Proof. At $(t,x,s)=(0,0,0)$ the derivative in s of the left side is a fixed nonzero multiple of the endomorphism Dolbeault Laplacian on Q_H, from $W^{k+1,2}$ to $W^{k-1,2}$. It is invertible by the fixed complementary elliptic estimate. The parameter implicit-function theorem gives σ. The linear x derivative vanishes at the origin: the linear curvature variation is contracted to an expression in d^*x and its adjoint by the Kähler identities, and x is harmonic. Taylor's theorem gives (14); elliptic regularity gives the indicated spatial norms. Uniqueness gives K-equivariance. □

Set

$$\Psi_t(x)=e^{\sigma(t,x)}\cdot(\bar\partial_0+a(x)),\qquad H_{t,x}=h_0e^{2\sigma(t,x)}.$$

The latter is the transported metric on the original holomorphic presentation when $x\in Z$. The determinant equation is separate: multiplication of a bundle metric by b_t solves it and leaves the induced endomorphism metric unchanged.

For a Hermitian multiplicity tuple $S=(S_i)$ write

$$S^0=\bigoplus_i I_{F_i}\otimes S_i-c(S)I,\qquad c(S)=\frac{\sum_i r_i\operatorname{tr}S_i}{R}. \tag{15}$$

Define the finite covector by

$$\langle m_t(x),S\rangle=\int_X\operatorname{tr}\big(S^0\mathcal{M}_t(\Psi_t(x))\big)dV_0. \tag{16}$$

At $x=0$, equivariance makes $\sigma(t,0)$ scalar on each multiplicity space, preserving the factor decomposition. Chern–Weil on each factor and the lift (15) give

$$m_t(0)=-\zeta(t),\qquad \rho_t=m_t-m_t(0). \tag{17}$$

The pairing in (16) is the ordinary multiplicity trace, not the spectral mass in (4).

2.3. **The quadratic ambient estimate.** Use the Hermitian L^2 pairing linear in the first variable, and $\omega_{\mathrm{flat}}(v,w) = \mathrm{Im}\langle v, w\rangle$. The affine space of unitary connections has its constant symplectic form determined by ω_t. Its pullback under Ψ_t is denoted Ω_t, with the normalization

$$d\langle \rho_t, S\rangle = 2\Omega_t(iC_x S, \cdot), \qquad \rho_t(0) = 0. \tag{18}$$

This is the gauge moment identity, using the K-equivariance of Ψ_t. The form is closed and invariant on the ambient ball; it need not have type $(1,1)$ for the finite complex structure.

Proposition 2.3. *In the canonical harmonic coordinates,*

$$\|\Omega_{t,x} - \omega_{\mathrm{flat}}\| \le C(|t| + \|x\|^2) \tag{19}$$

as real bilinear forms on the entire ambient representation.

Proof. It suffices to prove vanishing of the spatial first derivative at $(t,x) = (0,0)$. Write the homogeneous quadratic terms as

$$a(x) = x + a_2(x) + O(r^3), \qquad \sigma(0,x) = \sigma_2(x) + O(r^3).$$

Here

$$a_2(x) = -d^* G_2(x \wedge x) \in \mathrm{im}\, d^*.$$

For the action $g \cdot \bar\partial = g\bar\partial g^{-1}$, the $(0,1)$ component is

$$\Psi_0(x) - \bar\partial_0 = x + a_2(x) - d\sigma_2(x) + O_{C^j}(r^3). \tag{20}$$

Both displayed quadratic terms are orthogonal to harmonic one-forms. Their real parameter derivatives are also orthogonal to the entire harmonic space. Every term linear in x in the pullback of the constant affine form pairs a harmonic tangent with such a derivative, so vanishes. The remaining error is $O(r^2)$ in operator norm. Smooth dependence on t contributes $O(|t|)$. □

This is a statement about the canonical recursion, not all analytic coordinates tangent to the identity. A change $x \mapsto x + F_2(x) + \cdots$ can introduce a harmonic quadratic term, leaving only an $O(r)$ estimate in the new coordinates. The distinction will matter for the formality coordinates in Section 7.

2.4. **A positive form before the finite quotient.** For $x, v \in V_{\mathrm{det}}$ define

$$\mathcal{B}_{t,x}(v,v) = \inf_{\eta \in Q_0 W^{1,2}} \|J_x v - D_a \eta\|^2_{H_{t,x},\omega_t}. \tag{21}$$

The fixed estimate $\|\eta\|_{W^{1,2}} \le C\|d\eta\|$ on Q_0 and multiplication by $a = O(r)$ give, by absorption,

$$\|\eta\|_{W^{1,2}} \le C\|D_a\eta\|_{H_{t,x},\omega_t}. \tag{22}$$

Thus the normal equation for this minimum has a uniformly coercive operator on the fixed Hilbert space $Q_0 W^{1,2}$.

Proposition 2.4. *The form $\mathcal{B}_{t,x}$ is smooth, Hermitian and positive on a common neighborhood, and*

$$(1 - C\delta)\|v\|^2 \le \mathcal{B}_{t,x}(v,v) \le (1 + C\delta)\|v\|^2, \qquad \delta = |t| + r^2. \tag{23}$$

For a genuine integrable tangent v the full gauge minimum is

$$\inf_{\gamma \in W^{1,2}(\mathrm{End}_0 E_0)} \|J_x v - D_a \gamma\|^2 = \inf_{\xi \in \mathfrak{g}_0} \mathcal{B}_{t,x}(v - C_x\xi, v - C_x\xi). \tag{24}$$

All norms in the minima use $H_{t,x}, \omega_t$.

Proof. First let $t = 0$. Since $H_1 J_x = I$, harmonic orthogonality gives

$$\|J_x v\|^2 = \|v\|^2 + O(r^2)\|v\|^2.$$

The metric $H_{0,x}$ changes squared endomorphism norms by $1 + O(r^2)$. In the reference norm, $d^* J_x v = 0$, so

$$|\langle J_x v, D_a \eta\rangle_{H_{0,x},\omega_0}| \le Cr\|v\|\|\eta\|_{W^{1,2}}.$$

Indeed the $d\eta$ term is zero before metric variation, the $[a, \eta]$ term is $O(r)$, and the metric variation contributes $O(r^2)$. Coercivity bounds the minimizer by $Cr\|v\|$ in $W^{1,2}$. The removed energy is at most $Cr^2\|v\|^2$. This proves (23) at $t = 0$. Smooth dependence of the uniformly coercive inverse contributes $O(|t|)$. The inverse also proves smoothness of $\mathcal{B}$.

Every gauge zero-form splits uniquely as $\xi + \eta$, with $\xi \in P_0$ and $\eta \in Q_0 W^{1,2}$. Substitute (11), minimize over η, and replace ξ by $-\xi$ to obtain (24). For an integrable tangent, differentiating the full integrability equation gives $D_a J_x v = 0$. Hodge minimization therefore identifies the left side with the harmonic Kodaira–Spencer squared norm. □

The same construction on all of V uses the full complement of P, including gapped scalar zero-forms. On V_{det} it agrees with (21), since trace and trace-free forms are orthogonal and preserved by D_a. No inverse of $C_x^* C_x$ or $D\kappa_x$ enters the proof.

3. Finite balancing and Hermitian–Einstein metrics

3.1. **The finite correction.** We recall the finite-dimensional result used below. Its independence from weak orbit directions is the reason that the ambient estimate of Proposition 2.3 suffices.

Lemma 3.1 ([Wue26h, Theorem 2.1])**.** *Let V be a finite quiver representation with compact group $K = \prod_i \mathrm{U}(W_i)$ and flat moment map μ as in (3). On a ball suppose Ω is closed and invariant, $\rho(0) = 0$,*

$$d\langle \rho, S\rangle = 2\Omega(iC_x S, \cdot), \qquad \|\Omega - \omega_{\mathrm{flat}}\| \le \varepsilon.$$

For sufficiently small ε, depending only on the fixed multiplicity dimensions, a point with $\mu(x) = \zeta$ has a correction $y = e^S x$ with $\rho(y) = \zeta$ and $\|S\|_{\mathrm{op}} \le A\varepsilon$. Conversely, a point with $\rho(y) = \zeta$ has a correction of the same size to the flat level. Here ζ is central, and the starting point lies in a smaller ball so that the correction stays in the domain. Both assertions restrict to any locally G-invariant analytic subset. They identify the two local compact quotients continuously, including changes of stabilizer.

Proof. We recall the projection estimate and degree argument. For an orthogonal vertex projection tuple P, write $x = n + u$, with n preserving and u crossing the projected subspaces. On $u = 0$ the infinitesimal field of P vanishes, so both normalized moment components paired with P vanish. Integration along $n + su$ gives

$$|\langle \rho(x) - \mu(x), P\rangle| \le \varepsilon \|C_x P\|^2. \tag{25}$$

The integration is in the ambient representation; its intermediate points need not satisfy the obstruction equations.

Put $H = K_x$. Work in the Hermitian centralizer of H, perpendicular to its Hermitian stabilizer algebra:

$$\mathfrak{p}_H = \mathrm{Herm} \cap \mathrm{Cent}(H) \cap \mathfrak{h}^{\perp}.$$

The pairing here is ordinary multiplicity trace. On the connected H-fixed ball, the normalized moments annihilate $\mathfrak{h}$ and are invariant under H, hence lie in $\mathfrak{p}_H$. The central target belongs to this space if either moment attains it at the starting point.

Let $M = \sum_i m_i$. If $S \in \mathfrak{p}_H$ has operator norm $a > 0$, a spectral projection above a global eigenvalue gap of size at least $a/(M-1)$ satisfies $C_x P \neq 0$. Otherwise P and $I - P$ would fix x and be in $\mathfrak{h}$, while their trace pairings with S vanish, contradicting the strict spectral gap. The identity is in $\mathfrak{h}$, so $\operatorname{tr} S = 0$. If $\mathfrak{p}_H = 0$ no correction is needed; this also handles the trivial $M = 1$ action.

Diagonalize S. A crossing coefficient of squared norm w and eigenvalue difference d changes its squared norm by $e^{2d}w$. Consequently

$$\begin{aligned} &\langle \mu(e^S x) - \mu(x), P\rangle \geq 2e^{-4a}\frac{a}{M-1}\|C_x P\|^2, \\ &|\langle (\rho - \mu)(e^S x), P\rangle| \leq \varepsilon e^{4a}\|C_x P\|^2. \end{aligned} \tag{26}$$

For $a = 2M\varepsilon$ and sufficiently small ε, the first expression dominates. Thus the homotopy

$$\mu(e^S x) - \zeta + s(\rho - \mu)(e^S x), \qquad 0 \leq s \leq 1,$$

has no zero on the boundary of that ball in $\mathfrak{p}_H$. At $s = 0$ its only zero is $S = 0$, with positive derivative, since

$$\frac{d}{d\tau}\langle \mu(e^{\tau T}x), T\rangle = 2\|C_{e^{\tau T}x} T\|^2.$$

Its degree is one. The derivative is inverted only at this fixed point to define degree; no uniform lower eigenvalue bound is used. This proves forward correction. Starting from $\rho(y) = \zeta$, use the homotopy $\mu(e^S y) - \mu(y) - s(\rho - \mu)(y)$ and the same estimate for reverse correction.

The corrections stay in any locally invariant analytic subset. For uniqueness on compact quotients, polar decomposition reduces two representatives in the same complex orbit to y and $e^T y$. The squared norm along $e^{sT}y$ is convex, so the path stays in the ambient ball. Positivity gives

$$\frac{d}{ds}\langle \rho(e^{sT}y), T\rangle \geq 2(1-\varepsilon)\|C_{e^{sT}y} T\|^2.$$

The endpoint moment values agree by centrality; hence T fixes y. The same argument works for μ. Bounded correcting matrices, compactness of K, and uniqueness of the resulting classes give continuous inverse maps on nested local quotients. □

3.2. **Reconstruction of the metric.** The complementary equation makes $\mathcal{M}_t(\Psi_t(y))$ parallel. By (17), its vanishing is equivalent to

$$\rho_t(y) = \zeta(t).$$

Proposition 2.3 and Lemma 3.1 apply on the full representation V, then restrict to Z. For $x \neq 0$, use an ambient ball whose radius is a fixed multiple of $\|x\|$. Its form error is at most $C(|t| + \|x\|^2)$. We obtain

$$y = e^S x \in Z, \qquad \rho_t(y) = \zeta(t), \qquad \|S\|_{\mathrm{op}} \leq C\delta. \tag{27}$$

At $x = 0$, flat balance forces $\zeta = 0$; the normalized corrected moment also vanishes there, so take $S = 0$.

Use the trace-free parallel lift S^0 in (15). It acts on x as S does, preserves the determinant, and has norm at most a fixed multiple of $\|S\|$. The actual metric is

$$h^{\mathrm{HE}}_{t,x} = b_t(e^{S^0})^*\big(h_0 e^{2\sigma(t,y)}\big). \tag{28}$$

The star is pullback of a metric by the holomorphic isomorphism $e^{S^0} : E_x \to E_y$. The complementary estimate and $\|y\| \leq e^{2\|S\|}\|x\|$ give (6). Multiplication by b_t supplies the scalar determinant equation, hence the full HE equation.

3.3. **Coverage of nearby HYM solutions.** We specify the qualitative inputs and their compatibility with the quantitative construction. In the integrable HYM specialization of [Del25], Section 3.2 makes the subsolution condition automatic. Proposition 4.3 gives the convergent harmonic slice and local classification; Proposition 4.5 gives the complementary equation. Corollary 5.2 identifies nearby holomorphic morphisms with parallel intertwiners. Proposition 5.6 supplies a common-parameter inverse, Lemma 7.3 its open integrable image, and Theorem 7.4 the local quotient topology. These results permit the complete multiplicity algebra in (1).

Use a sufficiently small $W^{k,2}$ neighborhood of the central operator, with $W^{k+1,2}$ gauges. The usual elliptic slice argument gives the same maps in these Sobolev spaces; alternatively one may apply the cited smooth or Hölder slice at higher regularity and then use elliptic regularity. All actual reconstructed solutions are smooth. On the fixed Hermitian bundle h_0, the representative of a full solution is

$$\widehat{\Psi}_t(y) = b_t^{1/2} \cdot \Psi_t(y), \qquad y \in Z, \quad \rho_t(y) = \zeta(t). \tag{29}$$

Its determinant connection is the normalized HE determinant connection at ω_t, transported to $\det h_0$, and is independent of y. We use determinant-preserving unitary gauges, up to the ineffective constant scalar. Removing the known central gauge $b_t^{1/2}$ before taking the projective slice and restoring it afterward is essential: $\det h_0$ itself need not be HE for ω_t. The scalar correction is smooth in t and invisible on $\operatorname{End} E_0$.

For completeness, the passage from an arbitrary nearby positive gauge to our complementary one uses no varying small eigenvalue. If its logarithm is s, change the harmonic representative by a parallel e^h and consider

$$L(s,h) = \tfrac{1}{2}\log(e^{-h}e^{2s}e^{-h}). \tag{30}$$

The derivative in h of its parallel projection at the origin is $-I$. The finite implicit-function theorem therefore makes L complementary on a common neighborhood. Uniqueness in Lemma 2.2 identifies it with $\sigma(t,y)$ after unitary gauge. The local slice inverse cited above then identifies the actual solution with (29). Reverse finite balancing supplies a flat-balanced x in the same orbit.

Choose successive coefficient radii, for example $R_0/16, R_0/8, R_0/4, R_0/2$, inside the common ambient R_0 ball. After shrinking the parameter and actual neighborhoods, every construction stays inside its domain. This proves coverage of an open neighborhood of actual solutions, not only existence for a prescribed family. Parallel-intertwiner classification identifies isomorphic harmonic representatives with the same local G-orbit. Uniqueness of the unitary HE class and the finite uniqueness argument identify precisely the same equivalence classes. The common slice inverse and finite quotient continuity prove the local homeomorphism in Theorem 1.1.

3.4. **Smooth local pieces.** Fix a flat-balanced point x_0 and put $H = K_{x_0}$. Polar convexity gives $G_{x_0} = H^{\mathbb{C}}$. The correction S_0 in Lemma 3.1 lies in the centralizer of H, so $y_0 = e^{S_0}x_0$ has the same stabilizer. On the H-fixed subspace the map

$$(T,u) \longmapsto \rho_t(e^T e^{S_0}u) - \zeta(t), \qquad T \in \mathfrak{p}_H, \tag{31}$$

takes values in $\mathfrak{p}_H$. Its T derivative at $(0, x_0)$ has positive quadratic form $2\Omega_t(iC_{y_0}T, C_{y_0}T)$, hence is invertible at that point. The finite implicit-function theorem gives a smooth local correction. Its domain may shrink with the least positive orbit eigenvalue; the uniform estimate uses only the bounded value S_0, not the derivative of this correction.

The ambient holomorphic slice theorem [Sja95, Proposition 1.6 and Theorems 1.12, 1.20] applies to V at a central moment value. Indeed the symplectic form on the compact orbit is the pairing of ζ with Lie brackets, so the orbit is isotropic. This works also at nonzero irrational central values. Intersect the ambient slice with the reduced analytic space Z. In the unchanged-stabilizer locus, the quotient germ is represented by the H-fixed part of the transverse slice. That analytic subset can still be singular. Refine it into smooth analytic pieces and refine their boundaries by a local Whitney stratification. These pieces have smooth lifts; the compact slice and (31) give balanced lifts.

Repeating at the other stabilizers yields a local cover by such pieces, using the finite orbit types of the ambient compact linear action and a locally finite analytic refinement. Both quotient constructions use the same complex orbits and the same pieces. This proves their smooth identification. It neither identifies stabilizer type with smoothness nor asserts a uniform lower radius for these individual charts.

4. The complete small endomorphism spectrum

The spectral argument uses the exact equivariance identity (11), separately from the moment estimate. We first keep the reference metric, allowing nonintegrable ambient coefficients.

Proposition 4.1. *For every sufficiently small $x \in V_{\mathrm{det}}$, the self-adjoint zero-form operator $D_a^*D_a$ in the reference metric has exactly N eigenvalues below one fixed threshold $\gamma > 0$ and*

$$(1 - Cr^2)\nu_j(x) \le \lambda_j^0(x) \le (1 + Cr^2)\nu_j(x), \qquad 1 \le j \le N. \tag{32}$$

Its kernel is exactly $\ker C_x$ inside P_0, and the remaining spectrum is bounded below by γ. At an integrable point this is the endomorphism Dolbeault spectrum.

Proof. Put $A = [a, \cdot]$ and $D = d + A$. The decomposition of zero-forms is $P_0 \oplus Q_0L^2$. On P_0,

$$D\xi = -J_xC_x\xi, \qquad d^*D\xi = 0.$$

Let

$$G_f = P_0D^*DP_0, \qquad B_f = Q_0D^*DP_0, \qquad L_Q = Q_0D^*DQ_0.$$

Harmonic orthogonality and (10) imply

$$q_x \le G_f \le (1 + Cr^2)q_x, \qquad B_f = Q_0A^*DP_0, \qquad \|B_f\xi\|^2 \le Cr^2\langle G_f\xi, \xi\rangle. \tag{33}$$

The last inequality is relative to the energy of the same vector. An absolute bound by $Cr^4\|\xi\|^2$ would be insufficient.

By (22), L_Q has a fixed positive lower bound γ_0. Its form domain is $Q_0W^{1,2}$ and its operator domain is $Q_0W^{2,2}$. Its resolvent is uniformly bounded from L^2 to $W^{2,2}$ for $0 \le \lambda \le \gamma_0/2$. The finite Schur form is

$$S_\lambda = G_f - \lambda I - B_f^*(L_Q - \lambda)^{-1}B_f,$$

and (33) gives

$$(1 - Cr^2)G_f - \lambda I \le S_\lambda \le G_f - \lambda I. \tag{34}$$

Completion of the square on $P_0 \oplus Q_0W^{1,2}$ identifies the negative index of $D^*D - \lambda$ with that of S_λ. The triangular change is a bijection on this form domain, since the resolvent correction lies in $Q_0W^{2,2}$.

As $\|G_f\| = O(r^2)$, the form $S_{\gamma_0/2}$ is negative definite for small r. Hence exactly N eigenvalues lie below $\gamma = \gamma_0/2$. The two form inequalities in (34), negative-index counting and finite min–max give (32). This reasoning counts multiplicities and includes zeros; it never divides by ν_j.

If $C_x\xi = 0$, exact equivariance gives $D\xi = 0$. Conversely a kernel element has parallel part ξ with $q_x(\xi) = 0$ by (34) at $\lambda = 0$. Then $B_f\xi = 0$, and positivity of the complementary operator forces its complementary part to vanish. The kernel is therefore literally $\ker C_x \subset P_0$. □

This is the relative Schur mechanism used for repeated-factor extensions in [Wue26b, Section 5] and the full local model of [Wue26e, Section 5]. Here canonical harmonic equivariance supplies its hypotheses on a general background.

4.1. **Transfer to the actual HE metric.** Let $y = gx$ be the correction in (27), with $g = e^S$. Conjugation $T_g\xi = g\xi g^{-1}$ preserves the complex rank-weighted trace-free algebra and satisfies

$$C_y T_g \xi = g(C_x\xi)g^{-1}.$$

Both energy and mass are therefore bounded in both directions by factors $e^{\pm 4\|S\|}$, and generalized min–max gives

$$e^{-8\|S\|}\nu_j(x) \le \nu_j(y) \le e^{8\|S\|}\nu_j(x). \tag{35}$$

The spectral algebra is complex. There is no requirement that nonunitary conjugation preserve its Hermitian real subspace.

Changing from h_0 to $H_{t,y}$ changes squared endomorphism norms by $1 + O(\delta)$. Changing ω_0 to ω_t changes both zero-form masses and one-form energies by $1 + O(|t|)$. Their common form domain is $W^{1,2}$, and the differential is unchanged. The Rayleigh quotient and min–max therefore transfer all ordered eigenvalues relatively. The scalar determinant correction has no effect on endomorphism norms. Finally, holomorphic pullback of both bundle and metric in (28) is an exact isometry intertwining the Laplacians. Proposition 4.1 and (35) prove (7), with a common complementary gap after decreasing the cutoff. The kernel identification is transported by the same maps.

For every induced endomorphism metric the splitting

$$\operatorname{End} E_x = \mathbb{C}I \oplus \operatorname{End}_0 E_x$$

is orthogonal and preserved by the differential and its adjoint. On the scalar part $D_a(fI) = (\bar\partial f)I$. The scalar Dolbeault Laplacian on the compact connected (X, ω_t) has a one-dimensional constant kernel and a uniformly positive first nonzero eigenvalue for small t. Taking the smaller of the scalar and trace-free cutoffs completes the full endomorphism statement of Theorem 1.1.

5. Harmonic quotient metrics and varying backgrounds

5.1. **The quotient comparison.** We now prove (8). Let v be a tangent to one of the smooth pieces, represented at a flat-balanced point x. Section 3.4 gives a locally smooth correction $y = g(x)x$. At the point under consideration choose its value $g = e^S$ with $\|S\| \le C\delta$. Differentiating this change of presentation gives gv plus a complex orbit vector. The latter is removed by the finite minimum. Similarly, a varying holomorphic gauge in the infinite-dimensional presentation contributes an exact D_a term, removed by the harmonic minimum.

More explicitly, Proposition 2.4 at y gives

$$\inf_{\gamma} \|J_y w - D_a\gamma\|^2 = \inf_{\xi\in\mathfrak{g}_0} \mathcal{B}_{t,y}(w - C_y\xi, w - C_y\xi).$$

The bounded value of g sends finite orbit directions bijectively to those at y, with norm distortion $1+O(\delta)$. Apply (23) before taking the infimum, and then set $w = gv$ modulo the vertical term. This proves

$$(1-C\delta)\inf_{\xi}\|v-C_x\xi\|^2 \le \|[v]\|^2_{L^2,t} \le (1+C\delta)\inf_{\xi}\|v-C_x\xi\|^2.$$

No estimate for Dg is required. A large derivative of a local balancing gauge near a stabilizer change thus does not affect the tensor constant.

At a flat-balanced point, differentiating $\mu = \zeta$ makes a balanced tangent perpendicular to Hermitian orbit directions. Centrality of ζ makes Hermitian and unitary orbit directions perpendicular. Hence the full complex-orbit minimum agrees with the compact Kähler quotient metric on such tangents. Positivity of $\mathcal{B}$ also identifies the infinitesimal kernel on integrable vectors with $\operatorname{im} C_x$. Together with the actual coverage and smooth pieces already proved, this completes Theorem 1.1.

The order of minimization is the one in [Wue26d, Section 8]: remove the uniformly coercive infinite-dimensional directions before minimizing over the possibly singular finite orbit. Taking an inverse of the finite orbit Hessian first would lose the stated uniformity. There is also no assertion about arbitrary untransported tangent limits. For the weight-one action on $\mathbb{C}$, for example, the quotient seminorm of $v = 1$ is zero at $x \neq 0$ and one at $x = 0$.

5.2. **Moving harmonic backgrounds.** The additive parameter error in Theorem 1.1 can be removed by choosing the finite model at the current polarization.

Proposition 5.1. *Keep the holomorphic factors F_i fixed and let ω_t vary near ω_0. There are smooth determinant-normalized HE metrics $h_i(t)$ with $h_i(0) = h_{i0}$. Form*

$$h_{\mathrm{gr},t} = \bigoplus_i h_i(t)\otimes I_{W_i},$$

and use its harmonic spaces V_t, norms, Green operators and full canonical obstruction κ_t. In these coordinates all conclusions of Theorem 1.1 hold uniformly with error $C\|x\|_t^2$. The reference metric is $h_{\mathrm{gr},t}$, the spectral mass is

$$\operatorname{Vol}(\omega_t)\sum_i r_i \operatorname{tr}(\xi_i^*\xi_i),$$

and the central target is still $\zeta(t)$. The factors may have unequal slopes for $t \neq 0$.

Proof. Each central factor is stable and simple. After separately solving its scalar determinant equation, its trace-free HE linearization is invertible. The parameter implicit-function theorem gives the smooth family $h_i(t)$. HE existence implies polystability, and simplicity of the fixed holomorphic factor then implies stability. This argument supplies nearby factor stability without assuming a uniform chamber-size statement.

The holomorphic Dolbeault complexes do not change, so all cohomology dimensions are fixed. Smooth elliptic perturbation gives smooth harmonic projections and Green operators, with uniformly positive complementary gaps on a smaller parameter neighborhood. Projection followed by the positive Gram square root gives smooth orthonormal bases in each factor Ext space. Tensoring with $\operatorname{Hom}(W_i, W_j)$ makes the trivialization G-equivariant. The zero-form kernel is the same multiplicity algebra: cross-Hom vanishing is a fixed holomorphic fact, and the scalar endomorphisms of each stable factor are parallel for every $h_i(t)$.

For this construction use the moving density dV_t and moment representative

$$\mathcal{M}_t^{\mathrm{mov}} = i\Lambda_{\omega_t}F^0, \qquad Q_{H,t} \text{ defined by the moving } L^2 \text{ product.} \tag{36}$$

At the split connection the mean curvature is a constant on each factor block, so $Q_{H,t}\mathcal{M}_t^{\mathrm{mov}}(0) = 0$ exactly. The fixed-density representative (12) would multiply this block endomorphism by dV_t/dV_0; when slopes differ that generally has a nonzero complementary projection. Nevertheless the integrated moment is unchanged, because

$$\mathcal{M}_t^{\mathrm{mov}} dV_t = iF^0 \wedge \frac{\omega_t^{n-1}}{(n-1)!}.$$

Thus the central target and rank conventions remain those of (3).

Identify the smoothly moving Hermitian complements with a fixed Sobolev space. The derivative of the complementary equation at $t = 0$ is invertible, and its small perturbation remains uniformly invertible. At unequal slopes it need not equal a scalar multiple of a Dolbeault Laplacian on the whole Hermitian endomorphism bundle; continuity of the actual derivative suffices. For each fixed t, the linear coefficient derivative of contracted curvature vanishes on a current harmonic one-form. This follows from the Chern-connection Kähler identities, which express the contraction in terms of $d_t^* x$ and its adjoint and do not require the entire mean curvature to be central. Consequently the complementary solution satisfies

$$\sigma(t,0) = 0, \qquad D_x\sigma(t,0) = 0, \qquad \|\sigma(t,x)\|_{C^j} \le C_j \|x\|_t^2. \tag{37}$$

Uniform bounds for second coefficient derivatives on a smaller product neighborhood give the same constant for all small t.

At every t, the quadratic Kuranishi term is current-coexact and the quadratic gauge term is current-exact. Both are orthogonal to the current harmonic space. The proof of Proposition 2.3 therefore gives $\Omega_{t,x} - \omega_{\mathrm{flat},t} = O(\|x\|_t^2)$. The complementary minimum has the same quadratic error, by Proposition 2.4 in the current norms. Parallel equivariance and $d_t^* J_{t,x} = 0$ give the same energy-relative Schur bound with uniformly controlled background gaps. Lemma 3.1 in the orthonormal quiver coordinates then gives $\|S\| \le C\|x\|_t^2$, and the actual metric, spectral and quotient transfers preserve that error. The product determinant is already HE, since the sum of its constant factor mean curvatures is constant.

Finally these moving coordinates retain common local coverage. Impose moving Coulomb gauge on an integrable operator using the complement of the unchanged holomorphic zero-form algebra. Its linearization is the moving positive zero-form Laplacian, uniformly invertible. The parameter slice theorem gives a common gauge neighborhood. For an integrable Coulomb representative a, Hodge decomposition gives

$$a = H_{1,t}a - d_t^* G_{2,t}(a \wedge a),$$

so contraction uniqueness identifies it with the moving canonical representative. Positive-gauge normalization and the reverse finite correction apply as before. Smooth changes between the unitary presentations for h_0 and $h_{\mathrm{gr},t}$ transfer nested neighborhoods. All quadratic comparisons remain in the specified moving harmonic coordinates; no arbitrary nonlinear coordinate change is used to preserve their order. □

6. A stable bundle with cubic deformation equations

We construct a compact locally free example for which the full obstruction in Theorem 1.1 is essential. The method is the curve-to-bundle construction of Thomas [Tho99], with a different bicubic and a triple-line fiber.

6.1. **The compact geometry and the Hilbert germ.** On $B = \mathbb{P}^2$ choose smooth cubic curves defined by R, P, Q, with pairwise transverse intersections and no common point. Such triples form a nonempty open set: choose transverse smooth P, Q, then a general smooth R avoiding their finite intersection and transverse to both. Choose $p \in \{P = Q = 0\}$, so $R(p) \neq 0$. In $\mathcal{P} = \mathbb{P}^2_{[x:y:z]} \times B$ put

$$X = \{x^3R + y^3P + z^3Q = 0\}, \qquad C = \{x = 0\} \times \{p\}. \tag{38}$$

Write ω_1, ω_2 for the restrictions of the two hyperplane classes.

Lemma 6.1. *The threefold X is smooth and connected, $K_X \simeq \mathcal{O}_X$, $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$, and $\operatorname{Pic} X = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$. The fiber over p is the triple line $x^3 = 0$.*

Proof. At a singular point the fiber derivatives imply $x^2R = y^2P = z^2Q = 0$. If all three fiber coordinates are nonzero, all three coefficient cubics vanish, a contradiction. If only one is nonzero, the corresponding smooth cubic has a nonzero base differential. If two are nonzero, the corresponding two cubics vanish transversely, and the nonzero coefficients multiplying their independent differentials cannot cancel. In both cases the total defining differential is nonzero. Thus X is smooth. Adjunction gives $K_X \simeq \mathcal{O}_X$. The hypersurface is ample, so Lefschetz gives the Picard assertion. The restriction sequence of the bicubic gives connectedness and the stated structure-sheaf cohomology. The fiber assertion follows from $P(p) = Q(p) = 0$ and $R(p) \neq 0$. □

Near p, the functions $U = Q/R$ and $V = P/R$ are analytic coordinates. The local equation is exactly

$$x^3 + Vy^3 + Uz^3 = 0. \tag{39}$$

Proposition 6.2. *The normal bundle of C in X is $\mathcal{O}_C(1) \oplus \mathcal{O}_C(-3)$, and its analytic Hilbert germ is*

$$\operatorname{Specan} \mathbb{C}\{a, b\}/(a^2b, ab^2). \tag{40}$$

Its tangent dimension is two. Every first-order direction extends to second order, whereas a direction with both coordinates nonzero is obstructed at order three.

Proof. The ambient normal bundle is $\mathcal{O}_C(1) \oplus \mathcal{O}_C \oplus \mathcal{O}_C$. The normal map to $\mathcal{O}_C(3)$, obtained from (39), is

$$(s, c, d) \longmapsto cz^3 + dy^3.$$

It is surjective, with kernel $\mathcal{O}_C(1) \oplus \mathcal{O}_C(-3)$. Since $H^1(N_{C/\mathcal{P}}) = 0$, the ambient Hilbert scheme is smooth of dimension four at C. The natural family of lines in nearby fibers has tangent isomorphism onto $H^0(N_{C/\mathcal{P}})$, so it gives the full analytic Hilbert chart, including infinitesimal deformations:

$$x = ay + bz, \qquad U = c, \qquad V = d.$$

Substitution in (39) gives the four coefficients

$$a^3 + d, \qquad 3a^2b, \qquad 3ab^2, \qquad b^3 + c \tag{41}$$

of y^3, y^2z, yz^2, z^3. Their vanishing is the exact relative Hilbert condition. Eliminate $d = -a^3$ and $c = -b^3$ to obtain (40).

The ideal has primary decomposition

$$(a^2b, ab^2) = (a) \cap (b) \cap (a, b)^3. \tag{42}$$

Its radical is (ab), and its nonzero minimal defining ideal lies in the cube of the maximal ideal. The embedding dimension is two. The order of this ideal is invariant under analytic changes of minimal coordinates, so the full germ cannot be defined by quadratic equations. In particular $a = b = \tau$ defines a deformation over $\mathbb{C}[\tau]/(\tau^3)$ but cannot extend to $\mathbb{C}[\tau]/(\tau^4)$ with the same first derivative: higher coefficients of a, b cannot cancel the leading cubic products. □

The reduced germ consists of two transverse smooth axes, with an embedded component at their intersection in the full analytic germ. Higher-order curve obstructions occur in other compact Calabi–Yau examples; for instance [ADDF04, Section 5.2] computes fifth-order initial equations for lines on the Fermat quintic. Here the cubic germ will be realized by a stable locally free bundle.

6.2. **The Serre bundle and its stability.** Set $L = \mathcal{O}_X(-2, 2)$. Its restriction to C equals $\det N_{C/X} = \mathcal{O}_C(-2)$. The ambient restriction sequence

$$0 \longrightarrow \mathcal{O}_{\mathcal{P}}(i-3, j-3) \longrightarrow \mathcal{O}_{\mathcal{P}}(i, j) \longrightarrow \mathcal{O}_X(i, j) \longrightarrow 0 \tag{43}$$

shows that both L and L^* are acyclic. For L, the ambient bundles are $\mathcal{O}_{\mathcal{P}}(-2, 2)$ and $\mathcal{O}_{\mathcal{P}}(-5, -1)$; for L^* they are $\mathcal{O}_{\mathcal{P}}(2, -2)$ and $\mathcal{O}_{\mathcal{P}}(-1, -5)$. Each is acyclic by the line-bundle cohomology of $\mathbb{P}^2$.

The Hartshorne–Serre construction [Arr07, Tho99] gives a rank-two locally free bundle with a section:

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{s} E \longrightarrow I_C L \longrightarrow 0. \tag{44}$$

Indeed the local extension generator is a nowhere-zero section of $\det N_{C/X} \otimes L^*|_C = \mathcal{O}_C$, and $H^1(L^*) = H^2(L^*) = 0$ gives its unique global lift, up to the usual scalar choice. This generator ensures local freeness along C; away from C one has an extension of line bundles. Put

$$A = E(1, -1), \qquad \det A = \mathcal{O}_X.$$

Since $L|_C = \mathcal{O}_C(-2)$, (44) and the restriction sequence give

$$h^0(E) = 1, \qquad H^1(E) = 0. \tag{45}$$

Proposition 6.3. *The bundles E and A are slope-stable for*

$$H_N = N\omega_1 + \omega_2, \qquad 1 < N < 1 + \sqrt{3}.$$

In particular A is stable of slope zero for the integral ample class $H = 2\omega_1 + \omega_2$.

Proof. The double dual of a saturated rank-one subsheaf of E is a line bundle with the same first Chern class, and its map to the locally free bundle extends across codimension two. By the Picard description, it is $\mathcal{O}_X(-k, -l)$ for some integers k, l. It therefore suffices to show that every twist $E(k, l)$ with a nonzero section has positive slope.

For all integers i, j the ambient group $H^1(\mathcal{O}_{\mathcal{P}}(i-3, j-3))$ vanishes. Thus

$$H^0(\mathcal{O}_X(i, j)) \neq 0 \quad \Longleftrightarrow \quad i, j \geq 0.$$

Moreover

$$H^0(I_C(i, j)) \neq 0 \quad \Longleftrightarrow \quad i, j \geq 0, \quad i + j \geq 1.$$

For $i \geq 1$, use a polynomial divisible by x; for $j \geq 1$, use a base polynomial vanishing at p. Constants do not vanish on C. Twisting (44) shows that a section of $E(k, l)$ can occur only in the union of the coordinatewise upper quadrants based at

$$(0, 0), \qquad (2, -1), \qquad (3, -2).$$

The mixed triple intersections are $\omega_1^2\omega_2 = \omega_1\omega_2^2 = 3$, and the pure cubes vanish. Writing degrees here without the common factor $1/2!$, one obtains

$$\begin{aligned} c_1(E) \cdot H_N^2 &= 6(N^2-1), \\ c_1(E(2,-1)) \cdot H_N^2 &= 6(2N+1), \\ c_1(E(3,-2)) \cdot H_N^2 &= 6(2+2N-N^2). \end{aligned} \tag{46}$$

All are positive on the stated interval. Adding a nonnegative twist increases degree since H_N is ample. Thus $\mu_{H_N}(E(k,l)) > 0$ whenever it has a section, equivalently $\mu_{H_N}(\mathcal{O}_X(-k,-l)) < \mu_{H_N}(E)$. This proves stability. Twisting preserves stability and gives the result for A. □

The third quadrant is necessary for this proof. It accounts for sections of $I_C(i,0)$ with $i \geq 1$. The argument proves the sufficient interval above independently of any broader stability assertion for a different bicubic.

6.3. **Equality of the bundle and curve deformation germs.**

Theorem 6.4. *The fixed-X, fixed-determinant analytic deformation germ of A is (40), including its nonreduced structure. In particular* $\dim \operatorname{Ext}^1(A,A) = 2$*, the quadratic Kuranishi obstruction vanishes, and the cubic obstruction is nonzero in minimal coordinates. Each reduced axis gives a genuine one-parameter family of pairwise nonisomorphic stable locally free bundles near A.*

Proof. We compare the deformation functors, not only their tangent dimensions. Let T be a local Artin $\mathbb{C}$-algebra. Every deformation of E over $X \times \operatorname{Spec} T$ has a section lifting s: induction over small extensions places the section-lifting obstruction in $H^1(E)$ tensored with the square-zero kernel, which vanishes by (45). The same induction and $h^0(E) = 1$ show that its module of sections is free of rank one. The lifted section is unique up to a unit of T. Since its closed-fiber zero is a regular codimension-two zero, its zero scheme is a flat local complete intersection deformation of C. Its determinant remains L, as $H^1(X, \mathcal{O}_X) = 0$ and the determinant is fixed.

Conversely, every embedded deformation C_T belongs to the ambient line chart used in Proposition 6.2. It is relatively $\mathbb{P}^1$, and its relative normal determinant equals $L|_{C_T}$ by adjunction: K_X is trivial and $\mathcal{O}_{\mathcal{P}}(-2,2)|_{C_T}$ has relative degree -2. The possible line from the local base is trivial. The fixed acyclicity of L^* persists after tensoring with T. The relative Serre construction therefore extends the local normal generator uniquely to a vector bundle E_T with regular section. Changing that generator by a unit only rescales the section and gives an isomorphic bundle. These two constructions are inverse on isomorphism classes and commute with base change.

Twisting by $\mathcal{O}_X(1,-1)$ preserves the deformation functor. Stability makes A simple; scalar automorphisms act trivially on its deformation base. The completed local ring is therefore the completion of (40). The constructions also apply to small analytic families: vanishing of $H^1(E)$, the one-dimensional space of sections, analytic base change and regularity of the zero scheme give inverse analytic germ maps. Thus the assertion includes the convergent structure and nilpotents, not only the completed ring.

The defining ideal has order three in minimal embedding dimension two. Hence a minimal Kuranishi map has zero quadratic part and nonzero cubic part. The two reduced axes give distinct points of the simple-bundle deformation germ. Stability and local freeness persist in a small neighborhood. □

Thomas's corrected example [Tho99, Addendum] has ring $\mathbb{C}[\varepsilon,\eta]/(\varepsilon^2,\eta^2)$. The triple-line geometry (38) instead gives a one-dimensional germ with cubic initial ideal. The use of the Serre method is common to both constructions.

6.4. **The full repeated-factor application.** Choose a Kähler form in $H = 2\omega_1 + \omega_2$ and the normalized HE metric on A. Set $E_0 = A \otimes \mathbb{C}^3$. Then

$$V = \operatorname{Ext}^1(A,A) \otimes \operatorname{End}\mathbb{C}^3, \qquad \dim_{\mathbb{C}} V = 18, \qquad N = 8.$$

Since $H^1(X,\mathcal{O}_X) = 0$, the determinant condition removes no degree-one harmonic direction.

Corollary 6.5. *The full local HYM problem at $A\otimes\mathbb{C}^3$, with all eighteen harmonic coordinates and the complete convergent matrix obstruction, satisfies every conclusion of Theorem 1.1. At fixed polarization the relative error is $O(\|x\|^2)$. There are eight small trace-free endomorphism eigenvalues, including their exact changing kernel, a uniform complementary gap, and the same relative harmonic metric comparison on genuine smooth pieces. Every sufficiently nearby actual solution is represented.*

Proof. Lemmas and propositions above verify compact smooth Kähler geometry, local freeness, stability, determinant and the analytic deformation germ. Thus all background hypotheses of Theorem 1.1 hold. Ext additivity gives the eighteen coordinates and the full multiplicity algebra gives the eight trace-free zero-form directions. Apply the theorem to this entire family. Small polarization variations are covered as well, by the theorem or by Proposition 5.1 in its moving coordinates. □

The scalar slice $u \otimes I_3$ retains the cubic obstruction of A by the blockwise canonical recursion. Its quadratic obstruction is zero, so a quadratic truncation would admit false actual deformations. No claim identifies the complete matrix relations by substituting noncommuting matrices into the two scalar equations.

There is an explicit spectral family with arbitrary scale separation. Parametrize one reduced axis in canonical harmonic coordinates by

$$f(z) = ze + h(z), \qquad \|e\| = 1, \quad h(z) \perp e, \quad h(z) = O(z^2).$$

Use the harmonic linear coordinate along e as parameter. Holomorphic divided differences give

$$\|f(z) - f(w)\|^2 = |z-w|^2\big(1 + O((|z|+|w|)^2)\big). \tag{47}$$

Indeed $\|h(z)-h(w)\| \le C(|z|+|w|)|z-w|$, and the two summands are orthogonal. The estimate remains relative when $z-w$ is arbitrarily small.

For any three nearby parameters put $x = \operatorname{diag}(f(z_1), f(z_2), f(z_3))$. The harmonic reconstruction is blockwise, hence integrable. Every loop coefficient is diagonal and normal, so $\mu(x) = 0$. The bundle is $A_{z_1} \oplus A_{z_2} \oplus A_{z_3}$. On each off-diagonal matrix unit the mass is $2\operatorname{Vol}(\omega_0)$ and the commutator energy is $\|f(z_i) - f(z_j)\|^2$. Thus for every unordered pair the finite eigenvalue is

$$\nu_{ij} = \frac{\|f(z_i) - f(z_j)\|^2}{2\operatorname{Vol}(\omega_0)} \tag{48}$$

with complex multiplicity two. Distinct parameters give trace-free kernel dimension two; one double collision gives dimension four; a triple collision gives dimension eight. The corresponding compact stabilizers before removing the ineffective scalar are $\mathrm{U}(1)^3$, $\mathrm{U}(1)\times\mathrm{U}(2)$ and $\mathrm{U}(3)$.

Take $(z_1, z_2, z_3) = (0, s, s+\varepsilon)$, where $0 < \varepsilon \le s \to 0$. Equations (47)–(48) and Theorem 1.1 give two exact trace-free zeros and six positive eigenvalues, with two entries at each scale

$$\frac{\varepsilon^2}{2\,\mathrm{Vol}(\omega_0)}(1+O(s^2)), \quad \frac{s^2}{2\,\mathrm{Vol}(\omega_0)}(1+O(s^2)), \quad \frac{(s+\varepsilon)^2}{2\,\mathrm{Vol}(\omega_0)}(1+O(s^2)). \tag{49}$$

This means comparison of ordered lists with repetitions, not exact equality of perturbed pairs. The constant is independent of ε/s, and the ninth trace-free eigenvalue has a fixed positive lower bound. The kernel dimensions at collisions are exact. The quotient metric conclusion holds on the full local family, with additional analytic refinement at the crossing of the reduced axes. The smooth-branch calculation in (49) tests relative uniformity; it does not assign a spectral exponent to the nilpotent structure.

7. Earlier cases and limits of the comparison

7.1. **Extension families.** For the bipartite two-step families of [Wue26a, Theorem 1.1], take $m_i = 1$ and restrict to the specified extension subspace. There are no composable products, so $a = x$ and $\kappa = 0$ on that subspace. The commutator form (4) becomes the weighted graph form. If σ is the total normalized balanced edge energy, then

$$\|x\|_t^2 = \mathrm{Vol}(\omega_t)\sigma.$$

Proposition 5.1 gives the earlier $O(\sigma)$ metric and spectral rates without assuming $|t| = O(\sigma)$. The statement is first made at the small balanced representative and then pulled back by the original harmonic identification and balancing gauge. Metric-order inequalities survive this pullback even when that original gauge diverges. The fixed extension, prescribed nonzero components and its proven stability cone remain the hypotheses for the earlier stability criterion. Tree determinant identities require their separate finite graph argument.

Allowing arbitrary fixed multiplicities gives the complete complex algebra of [Wue26b, Theorem 1.1 and Section 5], with dimension $\sum_i m_i^2 - 1$. The mass and energy agree after a common volume normalization. Simplicity and the fixed stable multiplicity orbit are still needed for an entirely positive trace-free small spectrum and a unique normalized metric. The present theorem also includes nonsimple polystable points; it does not give an HE metric to a nonsplit semistable bundle.

For the adjacent-arrow three-step family of [Wue26c, Section 3], the only quadratic product lies in the endpoint block. Every product involving that block vanishes. The complete canonical equation therefore terminates:

$$a = x - d_t^* G_{2,t}(x \wedge x).$$

Its integrability equations are precisely the displayed cup-product relations of that family. This is a consequence of its triangular algebra, not a truncation applicable to arbitrary quivers. The normalized harmonic energy is the sum of the balanced edge weights, so Proposition 5.1 preserves the earlier quadratic relative rate, including arbitrary positive ratios. The distinct line factors, adjacent-arrow subspace and stable connected supports remain part of the graph interpretation.

7.2. **Full local moduli and K3 coordinates.** The complementary-minimization result [Wue26d, Section 8] is an imported principle in a more general compact-complex formulation. The harmonic arguments here verify its hypotheses in the moving presentation. The 83-coordinate wall chart of that paper instead uses a fixed reference Hodge contraction. Its product geometry gives the required orthogonality in the current norms: each harmonic Hom block has one differential leg, and changing the polarization multiplies the pairing on

that block by a constant. Thus the reference-orthogonal correction remains orthogonal. Its quadratic quotient estimate is not obtained by treating an arbitrary change of analytic chart as an isometry. At a fixed endpoint the estimate has its endpoint-dependent constant C_l; no uniformity as $l \to 0$ follows from the fixed-background theorem. The additional endpoint and limiting-metric conclusions retain their original hypotheses and proofs.

The full model of [Wue26e, Sections 2, 4–6] is

$$E_0 = F_1 \oplus (F_2 \otimes \mathbb{C}^2) \oplus F_3 \oplus F_4 \quad \text{on } (\mathbb{P}^1)^3.$$

Its complete harmonic recursion has 119 coordinates, and its parallel trace-free algebra has dimension $1^2+2^2+1^2+1^2-1 = 6$. The line-factor product metrics are HE for the given product polarizations. The normalized energy q_t equals $\|x\|_t^2/\operatorname{Vol}(\omega_t)$, so the same $O(q_t)$ stationary metric, spectral and quotient rates follow. All six modes, including zeros, are retained. The general finite correction covers nonsimple flat-balanced points without a scalar-centralizer assumption. The explicit weak coefficient, geometric curve and parameter-dependent length assertions remain additional results of that example.

For the K3 comparisons in [Wue26g, Sections 4, 6] and [Wue26h, Section 3], one must distinguish canonical harmonic coordinates from the independently established convergent equivariant formality coordinates. Suppose F maps the quadratic locus to the full Kuranishi locus and $F(x) = x + O(\|x\|^2)$. Then $DF = I + O(r)$, and exact equivariance gives

$$DF_x C_x \xi = C_{F(x)} \xi. \tag{50}$$

Applying the derivative estimate to $C_x\xi$ itself yields a relative energy comparison even in a very weak orbit direction. The pullback ambient form has error $O(|t|+r)$, giving that rate in the straightened coordinates. If the verified symmetry makes F odd, then $DF = I + O(r^2)$, recovering $O(|t| + r^2)$. For the real equation one must also apply finite balancing to the pulled-back moment form. A nonlinear analytic coordinate isomorphism need not preserve a flat moment level. For quotient norms, (50) identifies the orbit directions, while correction derivatives disappear under minimization. The formality hypotheses, homogeneous rescaled limits and intrinsic distance results of those papers are retained, not inferred from a general obstructed germ.

The rank-three K3 example uses the opposite sign at its multiplicity-two vertex. In its notation $\mu_{13} = A^*A - BB^*$ and $\tau(t) = -28\pi t/3$. Equation (3) instead gives

$$\mu_1 = -\mu_{13}, \qquad \zeta_1(t) = 28\pi t/3 = -\tau(t).$$

The second scalar equation follows by trace. This identifies the same real level set with the curvature convention $+iF^0$.

For all these comparisons, common division of zero-form mass and one-form energy by $\operatorname{Vol}(\omega_t)$ leaves eigenvalues unchanged. The normalized graph convention uses edge weights $\|x_a\|_t^2/\operatorname{Vol}(\omega_t)$ and vertex masses r_i. The unnormalized convention uses $\|x_a\|_t^2$ and $\operatorname{Vol}(\omega_t)r_i$. No dimension-free statement is made.

7.3. **Boundaries.** The relative estimates require more than small absolute errors. For instance the nonnegative matrices

$$\operatorname{diag}(\varepsilon^2, r^2), \qquad \operatorname{diag}(\varepsilon^2 + r^4, r^2)$$

differ in norm by r^4, while the ratio of their least eigenvalues is $1 + r^4/\varepsilon^2$. It is unbounded when $\varepsilon/r^2 \to 0$. The moment projection estimate (25) and the spectral energy bound (33) prevent this failure in their respective arguments. Both use exact geometric identities.

The theorem is local near a fixed smooth locally free HE background with positive complementary elliptic gaps. It does not supply harmonic spaces or such gaps for a singular graded sheaf, or uniformly through collapse of the background geometry. The singular-flow obstruction [Wue26f, Theorem 1.1 and Corollary 1.2] has a precise different scope. It concerns a locally free rank-two extension on a compact Kähler surface,

$$0 \longrightarrow L \longrightarrow V \longrightarrow I_W M \longrightarrow 0, \qquad \deg L = \deg M,$$

where W is a nonempty zero-dimensional subscheme. Its induced line metric coefficient $u(T)$ along HYM flow satisfies $\sqrt{T}\,u(T) \to 0$ and $\log u(T) + \frac{1}{2}\log T \to -\infty$ on every nonempty compact subset away from $\operatorname{Supp} W$. This excludes the stated bounded metric-order and ordinary matrix-log comparisons with the categorical $T^{\pm 1/2}$ model for fixed compatible splittings. The stationary bubbling result in that work is separately restricted to its toric example. Neither assertion is a general impossibility theorem for stationary local reductions at singular sheaves.

Finally, stationary HE reconstruction and zero-form spectral estimates alone give no HYM trajectory, matching of a prescribed initial metric, asymptotic phase, differentiated response estimate, global metric completion or homogeneous blow-up limit. Those questions require additional dynamical or geometric arguments. The full analytic obstruction, genuine reduced tangent spaces and fixed-background dependence in Theorem 1.1 specify the local stationary result.

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